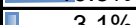


Type	Department	Discipline	Concept				
			Hours	Hourly rate	% hours	Cost	% cost
1	PM	Project manager	 206	£ 116.00	9.1%	 £ 23,896	 10.6%
3	Process	Process Engineering	 658	£ 105.00	29.1%	 £ 69,090	 30.8%
6	Civils	Civil/ Structural Engineering	 158	£ 105.00	7.0%	 £ 16,590	 7.4%
4	MEP	Mechanical Engineering	 202	£ 105.00	8.9%	 £ 21,210	 9.4%
5	E, C & I	E,I,C Engineering	 240	£ 105.00	10.6%	 £ 25,200	 11.2%
12	HVAC	HVAC Engineering	 64	£ 105.00	2.8%	 £ 6,720	 3.0%
6	DO - Civils	DO-Civil/ Structural	 156	£ 84.00	6.9%	 £ 13,104	 5.8%
4	DO - MEP	DO-Mechanical	 440	£ 84.00	19.5%	 £ 36,960	 16.5%
5	DO - E, C & I	DO-E,C & I	 82	£ 84.00	3.6%	 £ 6,888	 3.1%
2	Procurement	Procurement	 44	£ 105.00	1.9%	 £ 4,620	 2.1%
1	Admin	QA/ Admin	 8	£ 48.00	0.4%	 £ 384	 0.2%
			2258		100.0%	£ 224,662	100.0%
Third Party Costs						£ -	
Total including 3rd party						£ 224,662	

Discipline		Hours	Cost	% hours	% cost
Project management	1	 214	 £ 24,280	 9.5%	 10.8%
Procurement	2	 44	 £ 4,620	 1.9%	 2.1%
Process	3	 658	 £ 69,090	 29.1%	 30.8%
Mechanical/ piping	4	 642	 £ 58,170	 28.4%	 25.9%
E, C & I	5	 322	 £ 32,088	 14.3%	 14.3%
CSA	6	 314	 £ 29,694	 13.9%	 13.2%
EHS	7	0	£ -	0.0%	0.0%
HVAC	12	 64	 £ 6,720	 2.8%	 3.0%
Total		2258	£ 224,662	100.0%	100.0%

		Basic					
Type	Department	Discipline	Hours	Hourly rate	% hours	Cost	% cost
1	PM	Project manager	382	£ 116.00	13.7%	£ 44,312	15.9%
1	PM	Programme Manager	20	£ 116.00	0.7%	£ 2,320	0.8%
3	Process	Process Engineering	462	£ 105.00	16.6%	£ 48,510	17.4%
6	Civils	Civil/ Structural Engineering	254	£ 105.00	9.1%	£ 26,670	9.6%
4	MEP	Mechanical Engineering	304	£ 105.00	10.9%	£ 31,920	11.4%
5	E, C & I	E,I,C Engineering	316	£ 105.00	11.3%	£ 33,180	11.9%
12	HVAC	HVAC Engineering	52	£ 105.00	1.9%	£ 5,460	2.0%
6	DO - Civils	DO-Civil/ Structural	160	£ 84.00	5.7%	£ 13,440	4.8%
4	DO - MEP	DO-Mechanical	412	£ 84.00	14.8%	£ 34,608	12.4%
5	DO - E, C & I	DO-E,C & I	128	£ 84.00	4.6%	£ 10,752	3.9%
2	Procurement	Procurement	240	£ 105.00	8.6%	£ 25,200	9.0%
1	Admin	QA/ Admin	56	£ 48.00	2.0%	£ 2,688	1.0%
			2786		100%	£ 279,060	100%

Third Party Costs

£ 10,000

Total including 3rd party

£ 289,060

Discipline		Hours	Cost	% hours	% cost
Project management	1	458	£ 49,320	16.44%	17.67%
Procurement	2	240	£ 25,200	8.61%	9.03%
Process	3	462	£ 48,510	16.58%	17.38%
Mechanical/ piping	4	716	£ 66,528	25.70%	23.84%
E, C & I	5	444	£ 43,932	15.94%	15.74%
CSA	6	414	£ 40,110	14.86%	14.37%
HVAC	12	52	£ 5,460	1.87%	1.96%
Total		2786	£ 279,060	100.00%	100.00%

EN9802-Asymchem - Modification to existing footprint
FEED Cost Estimate (+30%)

Condition	Deliverables	No	Deliverable	Project manager	Process Engineering	Civil/ Structural Engineering	Mechanical Engineering	E,I,C Engineering	HVAC Engineering	DO-Civil/ Structural	DO-Mechanical	DO-E,C & I	Procurement	QA/ Admin	Third Party Costs	RBP comments	Hours	Cost	Type
TRUE		1.00	Project management																
FALSE		1.01	Kick off meeting	2	2	2	2	2									10	£ 1,072	1
FALSE		1.02	Project and progress Meetings - Internal weekly	12	12	12	12	12	6	6	12	6	2				92	£ 9,288	1
FALSE		1.03	Weekly progress meeting - Client meeting	8	8	8	8	8									40	£ 4,288	1
FALSE		1.04	Design reviews	12	12	12	12	12	6	6	12	6	2				92	£ 9,288	1
FALSE		1.05	Generate and maintain program	12													12	£ 1,392	1
FALSE		1.06	Project Activities	96	4	4	4	4	4					8			124	£ 13,620	1
FALSE		1.07	Documents review	8													8	£ 928	1
FALSE		1.08	Risk assessment/log	8													8	£ 928	1
FALSE		1.09	Programme update	24													24	£ 2,784	1
FALSE		1.10	PEP Project execution plan	8													8	£ 928	1
TRUE		2.00	Procurement																
FALSE		2.01	Cost estimate whole Project (+/-30%)	8	8								40				56	£ 5,968	2
FALSE		2.01	Procurement support										0				0	£ -	2
TRUE		3.00	Process																
FALSE		3.01	Process Design Basis		60												60	£ 6,300	3
FALSE		3.02	Process Description		32												32	£ 3,360	3
FALSE		3.03	Equipment list		16												16	£ 1,680	3
FALSE		3.04	Line list		24												24	£ 2,520	3
FALSE		3.05	Manual valve list		16												16	£ 1,680	3
FALSE		3.06	Auto valve list		16												16	£ 1,680	3
FALSE		3.07	Special Items List		16												16	£ 1,680	3
FALSE		3.08	Tie in point list		8												8	£ 840	3
FALSE		3.09	Relief device list		16												16	£ 1,680	3
FALSE		3.10	Instrument list		24												24	£ 2,520	3
FALSE		3.11	Equipment Datasheets		40												40	£ 4,200	3
FALSE		3.12	PFDs (Process Flow Diagrams)		16						16						32	£ 3,024	3
FALSE		3.13	Mass balance (including waste)		40												40	£ 4,200	3
FALSE		3.14	P&IDs & UIDs (outside Asymchem equipment scope)		80						80						160	£ 15,120	3
FALSE		3.15	Utilities capacity demand		40												40	£ 4,200	3
FALSE		3.16	HAC worksheet		24												24	£ 2,520	3
FALSE		3.17	HAC report		32												32	£ 3,360	3
FALSE		3.18	HAC drawings		8						40						48	£ 4,200	3
FALSE		3.19	Relief study		32												32	£ 3,360	3
FALSE		3.20	Design development		40												40	£ 4,200	3
TRUE		4.00	Mechanical/ Piping (MEP)																
FALSE		4.01	General equipment arrangement				16				32						48	£ 4,368	4
FALSE		4.02	Piping Isometrics with MTO (including tie-ins)				12				40						52	£ 4,620	4
FALSE		4.03	Piping & Valve MTO				8				12						20	£ 1,848	4
FALSE		4.04	Piping Classification (spec)		8		40										48	£ 5,040	4
FALSE		4.05	Piping main routing		8		24				40						72	£ 6,720	4
FALSE		4.06	3D perspectives of project	8		8	24	16			100						156	£ 14,368	4
FALSE		4.07	Disinvestment - mark ups			16	16				40						72	£ 6,720	4
FALSE		4.08	Utilities design specs (all utilities needed)				24	16									40	£ 4,200	4
FALSE		4.09	HVAC capacity demand						20								20	£ 2,100	4
FALSE		4.10	HVAC design specs		8				20		16						44	£ 4,284	4
FALSE		4.11	Room Data Sheet						8								8	£ 840	4
TRUE		5.00	E, C & I																
FALSE		5.01	Electrical Basis of Design input					4									4	£ 420	5
FALSE		5.02	Electric power consumer list (Load List)					8									8	£ 840	5
FALSE		5.03	Single Line Diagrams					4				4					8	£ 756	5
FALSE		5.04	Amtech Study - Cable sizing					16									16	£ 1,680	5
FALSE		5.05	Small Power Layout					4				16					20	£ 1,764	5
FALSE		5.06	Main cable routing (3D model)					8				36					44	£ 3,864	5
FALSE		5.07	IJB Layout - Allocation of Equipment items					12				6					18	£ 1,764	5
FALSE		5.08	Lighting Design and 2D layouts					16									16	£ 1,680	5
FALSE		5.09	Power Cable Schedules					4									4	£ 420	5
FALSE		5.10	Package interface Schedule					8									8	£ 840	5
FALSE		5.11	Containment BOM					6									6	£ 630	
FALSE		5.12	C & I installation specification					16									16	£ 1,680	5
FALSE		5.13	Instrument And Control Basis of design input					4									4	£ 420	5
FALSE		5.14	Instrument Block Diagrams					4				8					12	£ 1,092	5
FALSE		5.15	Control Cable Schedules					8									8	£ 840	5
FALSE		5.16	Electrical installation specification					16									16	£ 1,680	5
FALSE		5.17	I/O count (Schedule)		8			16									24	£ 2,520	5
FALSE		5.18	Explosion protection concept (initial ATEX Equipment Schedule)					16									16	£ 1,680	
TRUE		6.00	Civil/ structural (CSA)																5
FALSE		6.01	Scope of Work Civil			16											16	£ 1,680	6
FALSE		6.02	Preliminary floor plans and elevations			8				16							24	£ 2,184	6
FALSE		6.03	Demolition - mark ups and specification			16				8							24	£ 2,352	6
FALSE		6.04	Preliminary drawings for structural openings, drainage rainage, fixtures and fittings.			16				32							48	£ 4,368	
FALSE		6.05	Preliminary drainage plans - internal			8				16							24	£ 2,184	6
FALSE		6.06	Preliminary bund design for waste tank			8				16							24	£ 2,184	6
FALSE		6.07	Details of other architectural elements as necessary			8				16							24	£ 2,184	6
FALSE		6.08	Door elevations and details			8				16							24	£ 2,184	6
FALSE		6.09	Preliminary design of access platforms for equipment operation and maintenance (if required)			8				24							32	£ 2,856	
TOTAL Estimated Hours				206	658	158	202	240	64	156	440	82	44	8	£ -		2,258	£ 224,662	
Hours (%)				9.1%	29.1%	7.0%	8.9%	10.6%	2.8%	6.9%	19.5%	3.6%	1.9%	0.4%					
Rate				£ 116	£ 105	£ 105.00	£ 105.00	£ 105.00	£ 105.00	£ 84.00	£ 84.00	£ 84.00	£ 105.00	£ 48					
Person hour Cost projected				£ 23,896	£ 69,090	£ 16,590	£ 21,210	£ 25,200	£ 6,720	£ 13,104	£ 36,960	£ 6,888	£ 4,620	£ 384					

EN9802-Asymchem - Modification to existing footprint
Detail Design Cost Estimate (+30%)

Condition	Deliverables	No	Deliverable	Project manager	Process Engineering	Civil/ Structural Engineering	Mechanical Engineering	E,I,C Engineering	HVAC Engineering	DO-Civil/ Structural	DO-Mechanical	DO-E,C &I	Procurement	QA/ Admin	Third Party Costs	RBP comments	Hours	Cost	Type
TRUE		1.00	Project management																
FALSE		1.01	Kick off meeting	2	2	2											6	£ 652	1
FALSE		1.02	Project and progress Meetings - Internal weekly	20	20	20	20		10		20	10	10				150	£ 15,560	1
FALSE		1.03	Weekly progress meeting - Client meeting	20	20	20	20	20	10		20		10				140	£ 14,500	1
FALSE		1.04	Design reviews (3)	8	8	8	8	8		8	8	8					64	£ 6,304	1
FALSE		1.05	Generate and maintain program	20	4	4											28	£ 3,160	1
FALSE		1.06	Project Activities	240													240	£ 27,840	1
FALSE		1.07	Documents review	20													20	£ 2,320	1
FALSE		1.08	Risk assessment/loq	20													20	£ 2,320	1
FALSE		1.09	Programme update	20													20	£ 2,320	1
TRUE		2.00	Procurement																
FALSE		2.01	Procurement Operating Plan (In conceptual)										8				8	£ 840	2
FALSE		2.02	Client liaison, support, reporting										12				12	£ 1,260	2
FALSE		2.03	Vendor / Sub contractor selection and approval inc due diligence										16				16	£ 1,680	2
FALSE		2.04	NDA										8	16			24	£ 1,608	2
FALSE		2.05	Preparation and issue of fully specified and finalised tender documents RFQ - URS - SAQ - H&S - T&C URS prep' by DISC' ENG - For major items - Competitive Tender										16				16	£ 1,680	2
FALSE		2.06	Pre Tender Supplier Teams Meetings - DISC' ENG - Some PROC' SUPPORT										8				8	£ 840	2
FALSE		2.07	Expediting of Tender returns										8				8	£ 840	2
FALSE		2.08	Receipt of tenders, bid evaluation, commercial queries closed out										16				16	£ 1,680	2
FALSE		2.09	Completion of Commercial bid analysis										12				12	£ 1,260	2
FALSE		2.10	Identification and management of issues										4				4	£ 420	2
FALSE		2.11	Creation, maintenance and issue of tender schedule										4				4	£ 420	2
FALSE		2.12	QA/ admin support											40			40	£ 1,920	2
FALSE		2.13	Contract review, process queries, negotiate										40				40	£ 4,200	2
FALSE		2.14	Cost estimate whole Project (+/-10%)	8	8	8		8	8				40				80	£ 8,488	2
FALSE		2.15	LLI list and delivery time										20				20	£ 2,100	2
FALSE		2.16	Advise Asymchem on the potential Contractors to be invited for the contractor procurement process and perform a pre-qualification and shortlisting.										8				8	£ 840	2
TRUE		3.00	Process																
FALSE		3.01	Process Flow Diagrams		4												4	£ 420	3
FALSE		3.02	Equipment list		4												4	£ 420	3
FALSE		3.03	Line list		8												8	£ 840	3
FALSE		3.04	Manual valve list		4												4	£ 420	3
FALSE		3.05	Auto valve list		4												4	£ 420	3
FALSE		3.06	Special Items List		4												4	£ 420	3
FALSE		3.07	Tie in point list		8												8	£ 840	3
FALSE		3.08	Relief device list		2												2	£ 210	3
FALSE		3.09	Instrument list		4												4	£ 420	3
FALSE		3.10	Equipment Datasheets for procurement		16												16	£ 1,680	3
FALSE		3.11	Cause & Effect Matrix		16												16	£ 1,680	3
FALSE		3.12	Control philosophy		40												40	£ 4,200	3
FALSE		3.13	Relief Design		40												40	£ 4,200	3
FALSE		3.14	P&ID updates		24						32						56	£ 5,208	3
FALSE		3.15	Instrument datasheets		40												40	£ 4,200	3
FALSE		3.16	IO Schedule		4												4	£ 420	3
FALSE		3.17	HAZOP prep		8												8	£ 840	3
FALSE		3.18	HAZOP		40			40							10000	Assume External Chair - Teams	80	£ 8,400	3
FALSE		3.19	HAC updates		24						16						40	£ 3,864	3
FALSE		3.20	Emissions calculations		40												40	£ 4,200	3
TRUE		4.00	Mechanical/ Piping (MEP)																
FALSE		4.01	3D Modelling				24				80						104	£ 9,240	4
FALSE		4.02	MEP installation specification				40										40	£ 4,200	4
FALSE		4.03	General Equipment Arrangement		8		16				40						64	£ 5,880	4
FALSE		4.04	Piping Support Drawings & Schedule				24				40						64	£ 5,880	4
FALSE		4.05	Piping & Valve MTO				16				24						40	£ 3,696	4
FALSE		4.06	Tie-in Sketches		8		8				16						32	£ 3,024	4
FALSE		4.07	Pipe Support Drawings		4		16				40						60	£ 5,460	4
FALSE		4.08	Mechanical drawings equipment		8		40				16						64	£ 6,384	4
FALSE		4.09	Piping Classification		4		8				8						20	£ 1,932	4
FALSE		4.10	HVAC/Piping layout including demolition		8		8		8		24						48	£ 4,536	4
FALSE		4.11	Utilities design specs		24		16										40	£ 4,200	4
FALSE		4.12	Isometric drawings				8				24						32	£ 2,856	4
FALSE		4.13	Stress analysis				24										24	£ 2,520	4
FALSE		4.14	Layout design (floor plans, sections and elevations)		2		8				4						14	£ 1,386	4
FALSE		4.15	Room Data Sheet						8								8	£ 840	4
FALSE		4.16	HVAC design to tender pack						8								8	£ 840	4
TRUE		5.00	E, C & I																
FALSE		5.01	Updates to Electrical installation specification					12									12	£ 1,260	5
FALSE		5.02	Vendor Liaison + Updates to Package interface Schedule					8									8	£ 840	5
FALSE		5.03	Updates to Amtech Study including selectivity and protection settings					20									20	£ 2,100	5
FALSE		5.04	Update Lighting Design Including Emergency details					16				16					32	£ 3,024	5
FALSE		5.05	Updates to Power Cable schedules					8									8	£ 840	5
FALSE		5.06	IS Calculations					24									24	£ 2,520	5
FALSE		5.07	Instrument/control Cable Schedule					16									16	£ 1,680	5
FALSE		5.08	MCC Compartment schematics (typical)					4				8					12	£ 1,092	5
FALSE		5.09	Wall and floor transit details					12									12	£ 1,260	5
FALSE		5.10	Lightning Protection Design (Third Party Quote)					16									16	£ 1,680	5
FALSE		5.11	Update C&I Installation Specification					12									12	£ 1,260	5
FALSE		5.12	Electric power consumer list (Load Schedule)					8									8	£ 840	5
FALSE		5.13	Update Electric one-line diagrams (Single Line Diagrams)					4				8					12	£ 1,092	5
FALSE		5.14	MCC GA Update					4				6					10	£ 924	5
FALSE		5.15	Main cable routing (3D Coordination)					12				36					48	£ 4,284	5
FALSE		5.16	Scope of Work Automation (FDS)					12									12	£ 1,260	5
FALSE		5.17	Automation URS					12									12	£ 1,260	5
FALSE		5.18	I/O list (including card/rack allocation details)					16									16	£ 1,680	5
FALSE		5.19	Instruments sketches typical (instrument layout drawings)					4				24					28	£ 2,436	5
FALSE		5.20	BOM (containment+ Lighting)					12				12					24	£ 2,268	5
FALSE		5.21	Supplier Liason					8									8	£ 840	5
TRUE		6.00	Civil/ structural (CSA)																
FALSE		6.01	Plant Layout	4		8				8							20	£ 1,976	6
FALSE		6.02	Scope of Work Civil			4											4	£ 420	6
FALSE		6.03	Detailed existing floor plans and Elevations			16				24							40	£ 3,696	6
FALSE		6.04	Demolition - finalise mark ups			4				8							12	£ 1,092	6

FALSE		6.05	Detail drawings for structural openings Drainage, fixtures and fittings.			16				32							48	£	4,368	6					
FALSE		6.06	Finalise drainage plans - internal			8				16							24	£	2,184	6					
FALSE		6.07	Finalise bund design for waste tank			8				16							24	£	2,184	6					
FALSE		6.08	Details of other architectural elements as necessary			4				8							12	£	1,092	6					
FALSE		6.09	Door elevations and details			4				16							20	£	1,764	6					
FALSE		6.10	Design of access platforms for equipment operation and maintenance			8				16							24	£	2,184	6					
FALSE		6.11	Tender document generation			16				8							24	£	2,352	6					
FALSE		6.12	Tender queries			16											16	£	1,680	6					
FALSE		6.13	Construction and technical support & contract administration			80											80	£	8,400	6					
TOTAL Estimated Hours				382	462	254	304	316	52	160	412	128	240	56	£		10,000	2,258	£	289,060					
Hours (%)				16.9%	20.5%	11.2%	13.5%	14.0%	2.3%	7.1%	18.2%	5.7%	10.6%	2.5%											
Rate				£	116	£	105	£	105	£	84	£	84	£	84	£	105	£	48						
Person hour Cost projected				£	44,312	£	48,510	£	26,670	£	31,920	£	33,180	£	5,460	£	13,440	£	34,608	£	10,752	£	25,200	£	2,688.00